

Tutorials (SC1006)

Tut 5

Important concepts & methodology in this week

- When choosing which module(s) to connect
 - compatible in Electrical Signal level
 - compatible in Communication Protocol.
- Serial interfaces
 - Synchronous – has a separate clock signal to synchronize data transfer.
 - Asynchronous – has no clock signal. Data transfer is based local clocks at transmitter and receiver. Some sort of synchronization bits/pattern is typically used to align the two clocks.
- Determining the data transfer rate → choose the slowest module in the transfer chain.
- In general, if there is an even number of errors bits, parity checking fails.

Q5.1

(a)

Table 1a

No.	Module	Type (Input, Output or Bidirectional)
i.	Buttons	Input
ii.	Wifi	Bi-Directional
iii.	Touch Screen Controller	Input/Bi-Directional**
iv.	System Memory	Bi-Directional
v.	Display	Output

(b)

- i. Wifi.
 - a. WIFI0010AC
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor SPI interface.
- ii. Touch Screen Controller
 - a. TS002UART
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor UART interface.
- iii. System Memory
 - a. DRAM0002-16M16, SRAM0002-2M16, NOR0001-1M. Actual choice depends on other requirement such as memory speed (read/write), cost etc.
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor Parallel Bus interface.
 - c. 16 bit data bus compatible with Processor 16 bit parallel bus interface.

(c)

- Wifi. SPI Bus => 1-bit data in each direction (MOSI, MISO) => Serial Bus.
- Touch Screen Controller. UART interface => 1-bit data in each direction (TX, RX) => Serial Bus.
- System Memory. Parallel Bus => multiple data bits transferred simultaneously => Parallel Bus.

(d)

- Wifi – SPI interface. This interface has a common clock line (SCLK) to synchronise the data transfer. => synchronous
- Touch Screen Controller – UART interface. There is no common clock line between transmitter and receiver, all sync information are embedded in the data line signals. => Asynchronous.
- System Memory. Parallel Bus Interface. A Strobe signal is used to tell the receiver when to latch in the data on the data bus. => Synchronous.

(e)

Wifi module (WIFI0010AC)

- Max SPI Clock on Wifi Module = 50Mhz
- Max Processor SPI clock = 50Mhz
- => 50Mbps transfer rate (max).

(f) Only choose one example:

- DRAM (DRAM0002-16M16)
- Max data strobe rate supported by DRAM = 100Mhz
- Max Processor Parallel bus strobe rate = 200Mhz $\Rightarrow$ Max supported data strobe rate on this link = 100Mhz
- 16-bit interface => 16 bits data transferred with each strobe $\Rightarrow 100M * 16 = 1600Mbps$.

Q5.2

(a)

- 7-bits: 1001101 → 0x4D
- ASCII character → M

(b)

Even parity. Data+Parity bits → even number of '1's.

(c) No, the parity of the data sequence remains unchanged if two bits are erroneous. Hence, the parity check will fail.

(d)

(i) $T = 1/9600 = 0.104 \text{ ms}$.

(ii) 1 packet consists of Start, Data, Parity and Stop Bits.

9600 bits per second / 10 bits (1 packet) = 960 characters per second.

(e)

- 9600 bits per second / 9 bits (1 packet) = 1066 characters per second.
- Throughput is higher, but the integrity of the data packet cannot be ascertained.
- Which option to choose will depend on the signal line condition (noisy?) between the transmitter and receiver.

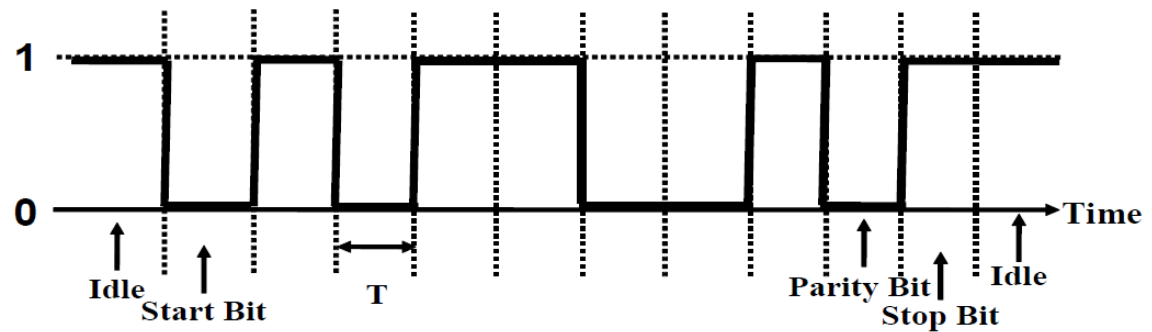


Fig 2

Q5.2

(f)

4800 baud rate: 7-bits: 10011012 → 0x4D

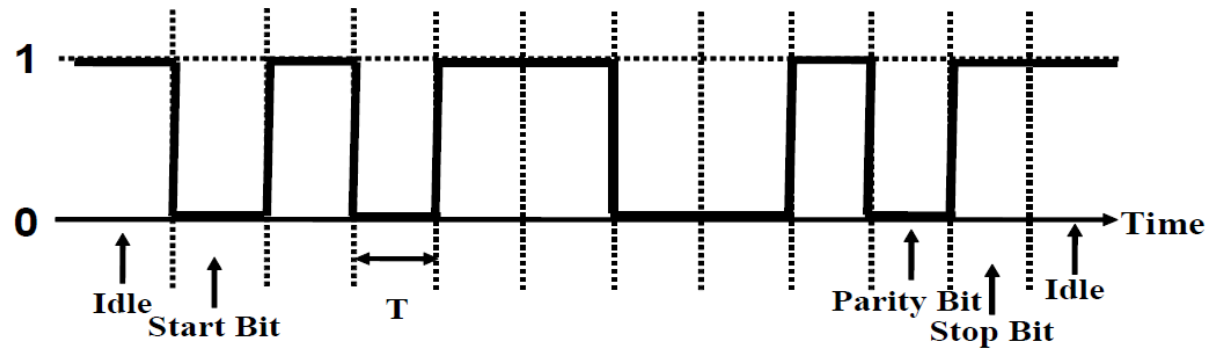


Fig 2

Original Frame									
S	D (lsb to msb)			P	E				
0	1011001			0	1				
Oversampled Frames (2x)									
Frame #1					Frame #2				
S	D (lsb to msb)			P	E	S	D (lsb to msb)	P	E
0	0110011			1	1	0	0001100	1	1

Frame #1: 0x66 → lowercase letter 'f'

Frame #2: 0x 18 → special character 'CAN'

Parity errors will be flagged for both frames (odd numbers of '1's received).